

ACCTG 211H — Mechanics Drill Set

Week 13 — Capital Budgeting: NPV, IRR, and Payback

Drill Set

Required: 1–5. Optional: 6–8. Complete the required items after instruction by **April 18, 2027, 11:59 p.m. Eastern**. Work individually by hand, check the key, and correct errors. Graded for completion; optional items are excluded from quizzes and the final.

1. (*Payback, even cash flows.*) A project costs \$240,000 and generates \$60,000 per year. Compute the payback period assuming cash arrives uniformly within each year.
2. (*Payback, uneven cash flows.*) A project costs \$100,000 and generates \$30,000, \$40,000, \$50,000, and \$60,000 in years 1 through 4. Compute the payback period assuming cash arrives uniformly within each year.
3. (*Present value of a single sum.*) A project returns a single cash flow of \$50,000 at the end of year 3. At a 10% discount rate, compute its present value.
4. (*NPV, annuity.*) A project costs \$75,000 and generates \$25,000 per year for four years. The discount rate is 10%. Compute the net present value.
5. (*NPV, uneven cash flows.*) A project costs \$60,000 and generates \$20,000, \$30,000, and \$25,000 in years 1 through 3. The discount rate is 8%. Compute the net present value.
6. (*Optional: IRR decision rule.*) Assume an initial outflow followed only by positive inflows. A project has an internal rate of return of 9% and the firm's cost of capital is 11%. Should the project be accepted? At the cost of capital, is the project's NPV positive, negative, or zero?
7. (*IRR, even cash flows.*) A project costs \$79,247 and returns \$25,000 per year for four years. Compute the annuity factor (cost $\div$ annual cash flow), then identify the IRR given that the four-year annuity factor equals 3.1699 at 10%.
8. (*Profitability index.*) A project requires an initial investment of \$50,000 and the present value of its future cash flows is \$58,000. Compute the profitability index and state the decision.

Supplied factors: 10%, year 3: 0.751315; 10%, four-year annuity: 3.169865. At 8%, years 1–3: 0.925926, 0.857339, 0.793832. Round final dollars; small factor-rounding differences are acceptable.